

Optimal cognitive distance and absorptive capacity

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Abstract

In this paper we test the relation between cognitive distance and innovation performance of firms engaged in technology-based alliances. The key finding is that the hypothesis of an inverted U-shaped effect of cognitive distance on innovation performance of firms is confirmed. Moreover, as expected, we found that the positive effect for firms is much higher when engaging in more radical, exploratory alliances than in more exploitative alliances. The effect of cumulative R&D turns out to be mixed. It appears to increase absorptive capacity, as expected, but there is clear evidence that it also reduces the effect of cognitive distance on novelty value, making it increasingly difficult to find additional novelty.

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1. Introduction

1.1. Learning and variety

The number of inter-firm alliances, especially those aimed at technological learning and new knowledge creation, has grown rapidly since the mid 1980s (Hagedoorn, 1993; Duysters and de Man, 2003). Resource interdependence and complementarities yield the most common explanation for the strong upsurge in the use of such inter-organizational ties (Richardson,

1972; Pfeffer and Nowak, 1976; Nohria and Garcia-Pont, 1991). This observation is in line with the basic premises of the resource-based view of the firm that has become one of the most prominent theoretical perspectives in strategic management (Wernerfelt, 1984; Barney, 1991; Teece et al., 1997; Rosenkopf and Nerkar, 2001; Ahuja and Katila, 2004). According to this view, which goes back to the work of Penrose (1959), firms differ in their resource positions and it is such resource heterogeneity that forms an important source of performance differences across firms. Strategic technology alliances enable firms to combine such heterogeneous resources (Porter, 1990; Prahalad and Hamel, 1990; Smith Ring and van de Ven, 1994; Hagedoorn and Schakenraad, 1994; Uzzi, 1997; Nooteboom, 1999; Ahuja, 2000b; Rowley et al., 2000). Particularly in high-tech sectors, alliances have become the cornerstone of innovation strategies of many companies. The majority of empirical

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studies have produced evidence that they positively affect corporate performance in terms of growth (Powell et al., 1996), speed of innovation (Hagedoorn, 1993) and organizational learning (Hamel, 1991; Ahuja, 2000a; Rowley et al., 2000; Hagedoorn and Duysters, 2002).

Recently, some studies have built further on these insights, focusing on where heterogeneous resources come from, how they can be accessed, and on the effectiveness of the various mechanisms that firms may employ (Rosenkopf and Nerkar, 2001; Rosenkopf and Almeida, 2003; Ahuja and Katila, 2004). These studies, however, tend to ignore an important question that goes beyond these issues; i.e. ‘how does such resource heterogeneity, once accessed, affect the inter-firm learning process?’, and the related question of ‘what are the implications for a firm’s innovation performance?’ The aim of this paper is to address these particular questions. Although this issue has already been addressed before by Mowery et al. (1996) who focus on the effect of technological overlap on interfirm knowledge transfer we complement this study in two important ways.

First, the authors focus only on the beneficial effects of small distances in cognition or in their terms, of a relatively large degree of technological overlap. Second, the authors focus on the process of interfirm knowledge transfer, but in this way abstract from the implications for firm performance. Both studies by Mowery et al. (1996, 1998) implicitly treat distance as only a problem instead of also an opportunity. The key contribution that our paper makes is that it shows the opposed effects of small versus large distances in cognition, and the implications of this combined effect for firm performance in terms of explorative versus exploitative innovation performance.

The main issue that we will address in this paper therefore concerns the differential performance effects when resources are either very similar, or alternatively, very different. Another central issue is how this outcome is affected by a firm’s R&D capabilities (technological capital). In other words, we aim to develop an understanding of the driving factors underlying the relation between resource heterogeneity and innovative performance.

1.2. Cognitive distance

In order to deal with this question effectively, we propose to interpret resource heterogeneity in terms of the *cognitive distance* between the firms that hold these different resources. Here, cognition denotes a broad range of mental activity, including proprioception, perception, sense making, categorization, inference, value judgments, emotions, and feelings, which all build on each other.

From the perspective that categories of cognition are constructed from action in the world, Nootboom (1992, 2000) inferred that to the extent that people have developed along different life paths and in different environments, they interpret, understand and evaluate the world differently. This leads to the notion of cognitive distance between people. Next, the question is how this notion of cognitive distance applies to firms. For organizations to achieve a common purpose, people do not have to agree on personal goals, and in the cognitive division of labor in a firm they will have dissimilar knowledge. However, they need to share certain basic perceptions and values to sufficiently align their competencies and motives. This requires a certain shared ‘interpretation system’ (Weick, 1979, 1995), ‘system of shared meanings’ (Smircich, 1983) or organizational ‘focus’ (Nootboom, 2000), established by means of shared fundamental categories of perception, interpretation and evaluation inculcated by organizational culture (Schein, 1985). Differences in such organizational focus yield cognitive distance between firms.

On the relation between cognitive distance and innovation performance, Nootboom (1992, 1999) proposed that there is an inverted-U shaped relationship. In first instance, as cognitive distance increases, it has a positive effect on learning by interaction. When people with different knowledge and perspectives interact, they stimulate and help each other to stretch their knowledge for the purpose of bridging and connecting diverse knowledge. This is related to Vygotsky’s (1962) notion of ‘Zoped’ (Zone of Proximal Development), where a tutor can ‘draw out’ cognitive development beyond what an agent could achieve alone.¹ Thus, cognitive distance yields opportunities for novel combinations of complementary resources. However, at a certain point cognitive distance becomes so large as to preclude sufficient mutual understanding needed to utilize those opportunities. Of course, a certain mutual understanding is needed for collaboration, and familiarity certainly breeds trust (Gulati, 1995a), which facilitates successful collaboration. However, too much familiarity may take out the innovative steam from collaboration. The challenge then is to find partners at sufficient cognitive distance to tell something new, but not so distant as to preclude mutual understanding. The curve can be reconstructed as the mathematical product of a line representing absorptive capacity (Cohen and Levinthal, 1990), which declines with cognitive distance, and a line representing the

¹ We thank a reviewer for pointing this relation to Vygotsky out to us.

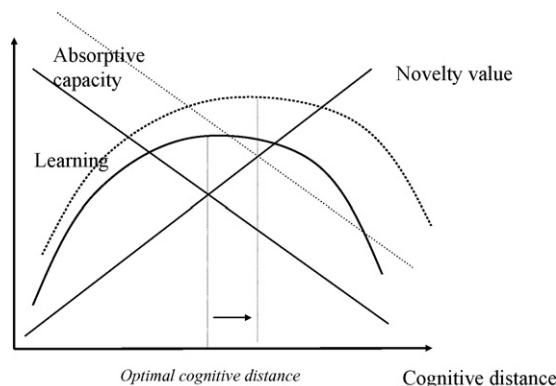


Fig. 1. Optimal cognitive distance.

novelty value of interaction, which increases with distance, as illustrated in Fig. 1.

1.3. Exploration and exploitation

We are also interested in the question of how far this inverted U-shaped relationship holds across different contexts. In particular, we distinguish between a context of exploration and a context of exploitation, following March's (1991) original argument that these two categories call upon different resources and capabilities held by firms. According to the logic of our argument, we expect the positive effect of cognitive distance (novelty value) to be higher when an innovation is more radical as is the case in exploration. We expect the positive effect of cognitive distance to be low(er) in collaboration processes that are geared towards exploitation. In collaboration purely for efficient production according to existing technology, distance is likely to primarily yield problems. Therefore, we anticipate a differential effect of cognitive distance on innovation performance depending on the extent of exploration versus exploitation.

The article proceeds as follows. First, in a theory section we discuss the underlying theory of knowledge and learning and we formulate our hypotheses. Then, we present details about the data, the specification of variables, and the estimation method. Next, we present our main findings and a discussion of the results. Finally, we provide the main conclusions and some indications for further research.

2. Theory and hypotheses

2.1. Variety and cognitive distance

In recent literature there is increasing consensus that resource heterogeneity provides a clear potential for

learning and innovation, and that strategic technology alliances can be considered as an efficient mechanism to effectuate this potential (Hagedoorn, 1993; Powell et al., 1996; Ahuja, 2000a; Rowley et al., 2000; Rosenkopf and Almeida, 2003). The theoretical framework underlying this growing understanding of the role of strategic alliances is formed by the resource-based view (RBV). The major focus in most (empirical) studies within this framework has been on comparing firms along industry and corporate characteristics. However, the majority of studies fail to adequately explain the underlying factors driving performance differences across firms (Hoopes et al., 2003). As a consequence, the basic question 'where do heterogeneous resources come from and how do they affect a firm's innovation performance?' has basically remained un-addressed so far. Some recent studies have started to shed some more light on these issues by focusing on the origins of heterogeneous resources, how they can be accessed, and the performance effects of the various search mechanisms employed for this purpose (Rosenkopf and Nerkar, 2001; Rosenkopf and Almeida, 2003; Ahuja and Katila, 2004). In other words, these studies have focused both on the *antecedents* of resource heterogeneity and their *consequences* for a firm's innovation performance. However, in-between creating access to heterogeneous resources and their ultimate effect on innovation lies unexplored territory, in the learning process between firms that starts when resources are brought together and subsequently combined. This recombination of resources leads to new knowledge creation. In order to fill this void, we focus on the causal factors that drive this learning process. Capturing these factors may enable us to understand the differential effects on innovation performance when resources are similar or alternatively, very different. For a further development of the notion of heterogeneity we propose to use the notion of *cognitive distance*. There are two reasons why in our opinion the use of cognitive distance is important in understanding learning processes.

A first reason relates to the social constructivist view of knowledge that underlies the notion of cognitive distance and that we will briefly summarize. According to this view, people that have been raised in different environments or conditions interpret, understand and evaluate the world differently (Berger and Luckmann, 1966; Nooteboom, 1992, 2000). From a firm's perspective, this implies that a firm's development along a specific path determines its organizational focus. The upshot of this is that to the extent that firms have developed in different technological environments, they operate at a certain cognitive distance, which provides

the basis for resource heterogeneity across firms. The notion of cognitive distance specifies causality and provides a stronger analytical grip and a clearer guide for empirical evaluation than the more general notion of resource heterogeneity. A second reason, as argued in Section 1, is that cognitive distance allows us to specify the role of absorptive capacity as decreasing, and novelty value as increasing, with increasing cognitive distance. In other words, an increase in cognitive distance creates both a problem and an opportunity.

This is in contrast with various bodies of innovation literature where distance is presented as only a problem instead of also an opportunity. In a study on alliance formation in the semi-conductor industry, Stuart (1998) argued that the most valuable alliances are those between firms with similar technological foci and/or operating in similar markets, whereas distant firms are inhibited from cooperating effectively. In a similar vein, the diversification literature argues that most is to be learned from alliance partners with related knowledge and skills (Tanriverdi and Venkatraman, 2005), or from areas that firms already possess capabilities in (Penner-Hahn and Myles Shaver, 2005). In the literature on international business also, a pervasive view is that cognitive distance is a problem to be overcome. Johanson and Vahlne (1977, 1990) employed the notion of ‘psychological distance’, which is seen as having an adverse effect on cross-cultural communication. When learning is discussed, in this strand of literature, it is mostly seen as learning to cope with transnational differences, by accumulating experience in cross-border collaboration (e.g. Barkema et al., 1997), rather than taking those differences as a potential source of learning to change home country products or practices. In sum, from the viewpoint of cognitive distance, the literature has focused on its negative effects and therefore overly stressed the benefits of homogeneous resources, thereby neglecting their negative effect (limited novelty value) and the positive effect of heterogeneous resources (large novelty value). So, cognitive distance provides us with a more complete understanding of the effects of heterogeneous resources.

The focus of this article is on the role of cognitive distance in strategic alliances, with a particular focus on new technology development and innovation. In this context we abstract from most of the different dimensions of cognition. We focus on the distance between alliance partners in terms of *technological knowledge*. Following the argument outlined above, such distance, in technological knowledge, among alliance partners yields an opportunity as well as a potential problem. The argument leads to the following hypothesis:

Hypothesis 1. In alliances, innovation performance is a parabolic, inverted-U shaped function of technological cognitive distance between alliance partners.

This is the core hypothesis that we want to test in this paper. In order to shed more light on this relationship we are interested in two issues. One is to test whether the expected relation between cognitive distance and innovation performance can be observed empirically. A second issue is, when such a relation can be observed empirically, to determine in how far this relation holds across different contexts. To investigate this, we make a distinction between exploration and exploitation. As we will argue, the role of cognitive distance differs profoundly between both contexts.

2.2. Exploration versus exploitation

The distinction between exploration and exploitation was discussed by March (1991). In short, exploitation is concerned with the refinement and extension of existing technologies, whereas exploration is the experimentation with new alternatives. Since exploitative and explorative learning are fundamentally different in nature, the role of cognitive distance is expected to have a different impact on the two types of learning.

Exploitation can be characterized as routinized learning, which adds to the existing knowledge base and competence set of firms without changing the basic nature of activities (Rowley et al., 2000; Hagedoorn and Duysters, 2002). This requires a strong mutual understanding of the firms involved, in order to coordinate rapidly and without errors. Cognitive distance creates uncertainty and complexity, which is undesirable in such a setting. Still, some cognitive distance may be needed in order to be able to make minor adaptations.

In contrast with exploitation, exploration can generally be characterized as a break with an existing dominant design and a shift away from existing rules, norms, routines, activities, etc., to allow for Schumpeterian novel combinations. This connects with the idea that firms have to move beyond local search by reaching for novel contexts in order to overcome the limitations of contextually localized search (Stuart and Podolny, 1996; Almeida and Kogut, 1999; Fleming, 2001; Rosenkopf and Nerkar, 2001; Rosenkopf and Almeida, 2003). By its very nature, moving beyond local search exploration is not about increasing efficiency of current activities. It is an uncertain process that deals with searching for new, technology based business opportunities (Nooteboom, 2000; Hagedoorn and Duysters, 2002). This requires access to and absorption of new insights and knowledge

that are, by definition, at a larger cognitive distance. Therefore, we hypothesize:

Hypothesis 2. In exploration, there is a stronger positive effect of cognitive distance on a firm's innovation performance than in exploitation.

2.3. Absorptive capacity

As argued, in the relationship between cognitive distance and innovation there are two opposing forces at work: a novelty effect that increases with larger cognitive distance and an absorption effect that decreases with larger cognitive distance. The novelty effect originates from making new combinations, and whether these combinations are potentially valuable is largely determined by the industry context. For the purpose of the present study, we consider this as largely exogenous, at least in the first approach. As will become clear later, the results force us to reconsider this assumption. On the other hand, we consider the absorption effect as more of an endogenous phenomenon that we need to include.

Following Cohen and Levinthal (1990), we see absorptive capacity as determined by R&D accumulating in technological capital (TC).² Firms with large(r) amounts of TC will generally show a better performance in dealing with cognitive distance, when compared to firms with small(er) amounts of TC. To further study this, we again differentiate between exploration and exploitation. Through R&D, firms can build up a patent portfolio, which can be seen as indicative of the codified knowledge that it has created. Moreover, R&D enables some of the important tacit skills that are needed to absorb the (codified and tacit) knowledge generated by its partners or potential others, who operate at a certain cognitive distance. Given the characteristics of exploration, with its focus on novelty, we argue that large amounts of technological capital strengthen the ability of firms to deal with a large(r) cognitive distance. This has a positive effect on its innovation performance. By contrast, given the characteristics of exploitation, with its focus on efficiency and incremental improvements, we do not expect that absorptive capacity through formalized RD is as important as in exploration. In exploitation, absorptive capacity may also be built up of more experience-based, tacit knowledge that resides within the firm and within relations with trusted, specialized suppliers and customers. Process innovations are generally not patented, for two

reasons. Such process knowledge tends to be more tacit, which makes it difficult to patent as this requires some level of codification. Moreover, there is less need to patent as the tacit nature of knowledge limits the potential for spillovers and enables to maintain (some) secrecy to competitors (Malerba and Breschi, 1997). Therefore, while R&D may still play a role for absorptive capacity in exploitation, this role is different and also less visible in a patent portfolio, when compared to exploration. So, we expect a limited effect of technological capital on the ability of firms to deal with cognitive distance in exploitation.

Therefore we hypothesize:

Hypothesis 3. In exploration there is a stronger positive effect of TC on innovation performance than in exploitation.

2.4. Mathematical specification

The hypotheses can be specified more precisely in mathematical terms, as follows:

The downward sloping line for absorptive capacity (A):

$$A = a_1 - a_2 CD; \quad a_1, a_2 > 0 \quad (1)$$

where CD is the (average) cognitive distance between the focal firm and its alliance partner(s).

The upward sloping line for novelty value (N):

$$N = b_1 + b_2 CD; \quad b_1, b_2 > 0 \quad (2)$$

Multiplying (1) and (2) results in the innovative performance (L) of a company:

$$L = AN = a_1 b_1 + (a_1 b_2 - b_1 a_2) CD - a_2 b_2 CD^2 \quad (3)$$

Eq. (3) specifies the basic model to be used for an econometric test.

For exploratory collaboration we expect the positive effect of distance (the slope b_2 in Eq. (2)) to be larger than for exploitative collaboration. It follows from Eq. (3) that *ceteris paribus*, i.e. for the same values, in exploration and exploitation, of the other parameters (a_1, a_2, b_1), for exploratory collaboration we expect a larger coefficient for both the linear and the quadratic term in distance. In collaboration only for efficient production, without any innovation, the novelty value of distance may be zero ($b_2 = 0$), and in that case collaborative performance only declines with distance.

From (3) it follows that optimal cognitive distance (CD^*) is:

$$CD^* = \frac{a_1 b_2 - b_1 a_2}{2 a_2 b_2} = \frac{1}{2} \left(\frac{a_1}{a_2} - \frac{b_1}{b_2} \right) \quad (4)$$

² Absorptive capacity is not only dependent on the external linkages of companies but also covers the ability of companies to exploit newly learned technology.

And optimal innovative performance (L^*) is:

$$L^* = a_1 b_1 + \frac{(a_1 b_2 - b_1 a_2)^2}{4 a_2 b_2} \quad (5)$$

From (4) it follows that for optimal distance to be positive, we must have $b_2 > b_1 a_2 / a_1$. From (4) and (5) it follows that for exploratory collaboration, where b_2 is larger, compared to exploitative collaboration, both optimal distance and innovative performance are larger. The proof of the latter proposition is given in [Appendix A](#).

One simple way to specify the effect of TC on absorptive capacity, in [Hypothesis 3](#), is to assume that higher TC yields an upward shift of the line representing absorptive capacity, i.e. a higher value of a_1 . Suppose that:

$$a_1 = c_1 + c_2 \text{TC}; \quad c_1, c_2 > 0 \quad (6)$$

Then, from (3) we find:

$$L = c_1 b_1 + c_2 b_1 \text{TC} + (c_1 b_2 - b_1 a_2) \text{CD} + c_2 b_2 \text{TC} \times \text{CD} - a_2 b_2 \text{CD}^2 \quad (7)$$

In other words, according to [Hypothesis 3](#) we then obtain a positive direct effect of TC and a positive interaction effect of TC and CD. This formula forms the basis for an empirical test of [Hypothesis 3](#).

3. Methods

3.1. Data

The hypotheses were tested on a dataset consisting of data for 116 companies in the chemicals, automotive and pharmaceutical industries. These companies were observed over a 12-year period, from 1986 until 1997. The panel is, however, unbalanced, because of new start-ups and mergers and acquisitions. Three types of data are combined in the empirical analysis. First, data about technology alliances were retrieved from the MERIT-CATI database. The 116 companies were selected to include the largest companies in these three industries that were also establishing technology based strategic alliances. Information on the establishment of alliances is hard to obtain for small or privately owned companies. Previous studies on inter-firm alliances also focused on leading companies in an industry ([Ahuja, 2000a; Gulati, 1995b; Gulati and Garguilo, 1999](#)). In total, 994 alliances were established in the period 1986–1996 among these companies.

Next, patent data were retrieved from the US Patent Office Database for all the companies in the sample, also those based outside the US. Working with U.S. patents – the largest patent market – is preferred over the use of

several national patent systems ‘... to maintain consistency, reliability and comparability, as patenting systems across nations differ in the application of standards, system of granting patents, and value of protection granted’ ([Ahuja, 2000a](#), p. 434). Especially in industries where companies operate on an international or global scale, U.S. patents may be a good proxy for companies’ world-wide innovative performance. We only include patents that have been successfully applied for, and the corresponding year is the year a company applied for a particular patent. For companies in the three sectors the financial data came from *Worldscope*, *COMPUSTAT* and data published on the companies’ websites.

3.2. Measures

The data yield a measure of cognitive distance (in technological capability), innovative output (patents), prior cumulative patents (as a proxy for technological capital and as a determinant of absorptive capacity), and requisite control variables (such as R&D expenditure, firm size, firm age, and year, country and industry effects). The dependent variable is measured in terms of the number of patents, with a distinction between more exploitative and more exploratory patents. Cognitive proximity (the inverse of distance) between firms is measured in terms of correlation between technological profiles derived from patent data. However, patent data are only available by firm per year, and cannot be attributed to individual alliance partners of the firm, so that the model has to be tested on the basis of average distance to the firm’s alliance partners.

An overview of the main variables, with their descriptions and average values, is provided in [Table 1](#).

3.3. Dependent variables

Yearly patent counts were used to derive the two dependent variables of exploitative and exploratory patents, as follows. For all companies technology profiles were created from the patents that a firm received in each patent class during the 5 years prior to a given year. A moving window of 5 years is the appropriate time frame for assessing technological impact ([Podolny and Stuart, 1995; Stuart and Podolny, 1996; Henderson and Cockburn, 1996; Ahuja, 2000a](#)). Studies about R&D depreciation ([Griliches, 1979, 1984](#)) suggest that knowledge capital depreciates sharply, losing most of its economic value within 5 years. This variable not only captures the effect of past R&D but also the internal mechanisms of knowledge production by internal interaction, since those also contribute to collective, firm

Table 1
Definitions of dependent and independent variables

Variable name	Variable description
Dependent variables	
Exploitative patents	Number of patents a firm successfully filed for in year t within patent classes in which the firm has been active in the 5 years prior to the given year
Explorative patents	Number of patents a firm successfully filed for in year t within patent classes in which it has not been active in the 5 years prior to the given year. The status of ‘explorative patent’ is kept for 3 years
Independent variables	
Cognitive distance	The average of the correlations between the focal firm’s technology profile and that of each of its alliance partners. The variable is transformed; the values range from 0 to 100, where increasing values stand for increasing distances between the technology portfolio of the focal firm and that of its alliance partners. Technology profiles are calculated based on the revealed technology advantage or specialization of each firm in each of the patent classes
(Cognitive distance) ²	Squared term of previous variable
Cumulative patents	Count of the number patents that a firm successfully filed for during the previous 5 years ($t - 5$ to $t - 1$). This variable represents the existing patent portfolio or technological capital of a firm
Control variables	
Age	The number of years since a company is founded
Firm size (ln revenues)	Natural logarithm of the total sales of the firm in $t - 1$ ($\times 1000$ Euro)
R&D intensity	R&D expenditures in $t - 1$ divided by total sales in $t - 1$
Year	Dummy variables indicating a particular year in the observed period 1986–1997
Chemical company	Dummy variable set to one if the firm is a chemical company (default = pharmaceutical company)
Car manufacturer	Dummy variable set to one if the firm is a car manufacturer (default = pharmaceutical company)
Europe	Dummy variable set to one if the firm is headquartered in Europe (default = Asian company)
US	Dummy variable set to one if the firm is headquartered in the U.S. (default = Asian company)

Note: All network variables are based on an alliance network representing all the technology-based alliances that were established in an industry during the 5 years prior to year t .

wide technological capital (Cohen and Levinthal, 1990). The patent classes were determined at two-digit level, which resulted in approximately 400 classes. The technology profiles enable us to make a distinction between exploitative and explorative technology classes. If a company successfully files for a patent in a patent class in which it has already filed for patents during the last 5 years, then it is considered to be an exploitative patent. A patent is labeled as explorative if a company successfully files for a patent in a patent class that is new for the firm, i.e. in which it did not have any patents during the last 5 years. Explorative patents keep this ‘status’ for 5 consecutive years. After that time this patent is considered to be exploitative. The 5-year period is fairly arbitrary and the alternative of a 3-year period was considered. Changing this time interval did not result in significant differences in the outcome of the empirical analysis.

The dependent variable ‘explorative patents’ was then made by adding up all the patents applied for in the year of observation in the explorative patent classes. The same was done for the variable ‘exploitative patents’, adding up the patents in all exploitative classes.

3.4. Explanatory variables

Cognitive distance was measured on the basis of CRTA, which is Pearson correlation index of the distribution across technological classes of the revealed technological advantages (RTA) of each firm relative to the other sample firms. The RTA of a firm in a particular technological field is given by the firm’s share in that field of the US patents granted to all companies in the study, relative to its overall share of all US patents granted to these companies. The RTA index varies around one, such that a value greater than one suggests that a firm is comparatively specialized in the technology in question, given its overall innovative performance. Positive values of CRTA indicate similarity of the pattern of relative technological specialization of firms, as it appears from the distribution of their patent activity across technological fields. For each firm and each year, a profile was constructed of its revealed technological advantage (RTA) in each patent class. A company’s RTA-index in a patent class is defined as the firm’s share of patents in that class (compared to all its alliance partners) divided by its share in all patent classes. The correlation coefficient was

computed pairwise between the RTA-profile of the focal firm and that of each of its alliance partners. The CRTA variable is then calculated as the average of these correlations. The values for CRTA can theoretically vary from -1 to 1 . As positive (negative) values indicate smaller (larger) cognitive distances, we choose to transform this variable into a new one (CD or ‘cognitive distance’) with a minimum value of zero and a maximum of one, where higher values indicate larger cognitive distance.

Cumulative technological capital is calculated as the number of patents that a firm obtained in the previous 5 years (as used in the calculation of the technology profiles, see above). Patents granted to a company are used to measure, in an indirect way, the technological competence of a company (Narin et al., 1987).

3.5. Control variables

In order to avoid bias from other effects on performance, control variables are included in the analyses. First, firm size is also expected to have an effect. We include the natural logarithm of ‘corporate sales’, a proxy for firm size, as a control variable. Firm size is expected to enhance exploitative learning (Acs and Audretsch, 1991). Large firms have the financial means and vast technological and other resources to invest heavily in R&D, and have advantages of scale (in specialization, capital outlays), diversification of risk, and a wider portfolio of activities into which unpredictable outcomes from R&D might fit. On the other hand, large size yields larger costs of coordination, a larger probability of original ideas being rejected in a multi-level hierarchy of evaluation and selection, less autonomy, less flexibility and less intrinsic motivation, more vested interests in existing products that new products might cannibalize. In large firms there is more systemic integrity to protect from disruption. For a survey of these arguments from the literature, see Nootboom (1994). Thus, we expect that larger firms have more resources to invest in R&D and to profit from its outcomes, while smaller firms tend to produce those outcomes more quickly, efficiently and radically. Nootboom (1991) hypothesized, and Nootboom and Vossen (1995) empirically confirmed that while in most industries large firms participate more in R&D than small firms, when small firms participate they do so more intensively and efficiently. Ahuja and Lampert (2001) also found that increasing firm size results in decreasing innovation performance. Also, larger firms are found to favor specialization along existing technological trajectories (Levinthal and March, 1993; Ahuja and Lampert, 2001). As a result, here we expect that while small firms have an overall disadvan-

tage, in terms of resources, this disadvantage is smaller in exploration than in exploitation.

Second, as documented extensively in the innovation literature, R&D expenditure is expected to have an effect on patents (Ahuja and Katila, 2001; Hagedoorn and Cloodt, 2003). Therefore, the natural logarithm of R&D expenditures is an interesting control variable but it is highly correlated with firm size. In order to separate R&D effects from size effects we chose to include R&D-intensity to take into account the variance in inputs for innovation activity (Hall and Ziedonis, 2001). We expect a positive and significant coefficient in the regressions.

Third, firm age may have an effect too. Generally, one would expect that organizational age is positively related to innovation, but that older firms show a greater tendency to build on older technology when compared to younger firms. Moreover, innovations of older firms are generally less influential than those of younger firms (Sorenson and Stuart, 2000). So, as organizations age, their competence to generate more innovations (or at least patents) improves, however, at a price of declining novelty value and importance of these innovations. Given these findings we predict that organizational age will have a positive effect on exploitation, which is essentially about refining and improving existing technology, whereas we expect a negative effect on exploration.

Finally, there may be unknown effects of specific years, for which use will be made of year dummy variables, and there may be regional effects, which we control by means of dummy variables for the EU and the US. Industry specific effects are also captured by two extra dummy variables. The propensity to patent may be partly determined by the nationality and/or the sector of the companies. Annual dummy variables may capture the ever-growing importance of intellectual capital, forcing companies to file more patents over the years, or macroeconomic conditions that may affect the three industries.

In sum, control variables are R&D intensity, firm size, firm age, and industry, regional and time effects. Statistics and correlations of variables are given in Table 2.

3.6. Estimation

The dependent variable is a count variable and takes only nonnegative integer values, i.e. the number of patents a firm filed for in a particular year. A Poisson regression approach provides a natural baseline model for such data (Hausman et al., 1984; Henderson and Cockburn, 1996). A theoretical argument for Poisson is that the hazard rate of achieving an innovation in time is Poisson distributed, and that this carries into the distur-

bance term. (Kamien and Schwartz, 1972; Loury, 1979; Dasgupta and Stiglitz, 1980; Lee and Wilde, 1980).

The basic Poisson model for event count data can be written as follows:

$$f(y_{ijt}|\lambda_{ijt}) = \frac{\exp(-\lambda_{ijt})\lambda_{ijt}^{y_{ijt}}}{y_{ijt}!} \quad (8)$$

where the parameter λ_{ijt} represents the mean and the variance of the event count and y_{ijt} the observed count variable.

It is furthermore assumed that:

$$\ln(\lambda_{ijt}) = \beta X_{ijt} \quad (9)$$

with X_{ijt} being a vector of independent variables and β a parameter vector.

The above specification in Eq. (8) assumes that the mean and variance of the event count are equal. However, for panel data on patent counts this assumption is likely to be violated, since it is well known that count data suffer from overdispersion (i.e. the variance exceeds the mean). This overdispersion is particularly relevant in the case of unobserved heterogeneity, i.e. the possibility that identical firms on the measured characteristics are still different on unmeasured characteristics. Since we use panel data with several observations on the same firms at different points in time, we modeled the data using a random effects Poisson estimator.³

Heckman and Borjas (1980) recommend a historical stock variable of the dependent variable as a control for unobserved heterogeneity. Along this line, Blundell et al. (1995) developed a pre-sample information approach in which unobserved heterogeneity is directly measured and entered into the model as a covariate. Blundell et al. (1995) argued that because the main source of unobserved heterogeneity in models of innovation lies in the different knowledge stocks with which firms enter a sample, a variable that approximates the build-up of firm knowledge at the time of entering the sample is a particularly good control for unobserved heterogeneity. Blundell et al. (1995) suggested that the pre-sample history of the dependent innovation variable is an appropriate proxy variable for a firm's knowledge stock upon entry into the sample. Ahuja and Katila (2001) used a

similar approach to control for unobserved heterogeneity in firm patenting. Unobserved heterogeneity may be the result of differences between companies in their innovation generating capabilities, and as a consequence, also in their propensity or ability to patent. Such unobserved heterogeneity, if present and not controlled for, can lead to overdispersion in the data or serial correlation. Therefore, in line with Hausman et al. (1984) we use a panel Poisson approach by introducing an individual unobserved effect in the conditional mean of the Poisson distribution. In particular, a gamma distribution with parameters θ_{ijt} and $1/\alpha$ is assumed for the conditional mean, where $\ln(\theta_{ijt}) = \alpha X_{ijt}$. This changes the resulting distribution of the dependent variable into a panel negative binomial the density of which equals:

$$f(y_{ijt}|\theta_{ijt}, \alpha) = \frac{\Gamma(\theta_{ijt} + y_{ijt})}{\Gamma(y_{ijt} + 1)\Gamma(\theta_{ijt})} \times \left(\frac{\alpha}{1 + \alpha}\right)^{y_{ijt}} \left(\frac{1}{1 + \alpha}\right)^{\theta_{ijt}} \quad (10)$$

where $\Gamma(\cdot)$ is a standard gamma distribution and $\alpha > 0$.

A negative binomial model has the advantage over the standard model that the former allows for a different mean and variance. More specifically, the ratio of the variance to the mean can be calculated from Eq. (10) as $1 + \alpha$, and is permitted to grow with the mean (Hausman et al., 1984).

4. Results

4.1. Cognitive distance

As a base case to compare our results against, we first present the outcome with only the control variables. Model 1 in Table 3 represents the impact of the control variables both on exploitative patents and explorative patents.

In Model 2, we present the results for the basic explanatory model, according to Eq. (7), which in addition to the control variables includes the linear and quadratic terms in cognitive distance and technological capability as explanatory variables. Technological capital – or the existing patent portfolio – t of a firm is supposed to improve its absorptive capacity improving the innovative (exploitative and explorative) performance for varying distances in cognition. As specified in Eq. (7), this is expected to yield a positive effect of TC as well as a positive effect of the interaction between TC and CD.

The results confirm the basic hypothesis (Hypothesis 1) that innovation performance is a parabolic, inverted-U

³ We ran a Hausman test for each regression in Table 3. This test is a formal way to determine if a random-effects model is misspecified and a fixed-effects model is necessary. The results show that the random effects model is consistent and is, thus, an appropriate specification. Fixed effects models give very similar results but we prefer not to use this specification since it does not allow for time invariant variables and it excludes firms that have not patented during the whole observation period, reducing the number of observations.

Table 3
Determinants of the patent rate of firms–1986–1997

Variable	Exploitative				Explorative			
	Model 1 Poisson	Model 2 Poisson	Model 3 Poisson	Model 3 Neg bin.	Model 1 Poisson	Model 2 Poisson	Model 3 Poisson	Model 2 Neg bin.
Explanatory variables								
Cognitive distance ^a		−0.9825 (0.6294)	1.1765 (0.8826)	2.5950 (2.1879)		6.7729*** (1.8050)	14.2109*** (2.4835)	8.4313** (3.8105)
(Cognitive distance) ²		0.0049 (0.0087)	−0.02535** (0.0120)	−0.0409** (0.0199)		−0.0827*** (0.0239)	−0.1778*** (0.0323)	−0.1152** (0.0503)
Cumulative patents /1000		0.2300*** (0.0491)	0.8827*** (0.1786)	1.3104** (0.5795)		0.8006*** (0.2076)	4.0631*** (0.7847)	2.4973** (1.2085)
((Cognitive distance) × (cumulative patents))/1000		−0.2700** (0.1131)	−3.7151*** (0.9136)	−5.0131* (2.9931)		−1.1482*** (0.4830)	−19.3650*** (3.9566)	−11.8095* (6.1742)
((Cognitive distance) ² × (cumulative patents))/1000			0.0446*** (0.0118)	0.0608* (0.0316)			0.2411*** (0.0500)	0.1519* (0.0789)
Control variables ^b								
Firm size (ln sales)	0.8269*** (0.0244)	0.7896*** (0.0247)	0.7807*** (0.0247)	0.2468*** (0.0541)	0.4272*** (0.0447)	0.3916*** (0.4516)	0.3824*** (0.0450)	0.1842*** (0.4690)
R&D-intensity	3.1899*** (0.2134)	2.5337*** (0.2119)	2.5167*** (0.2117)	−0.4191 (0.4110)	1.2763*** (0.2406)	1.1590*** (0.2409)	1.1465*** (0.2401)	0.6302** (0.2673)
Age	0.0139*** (0.0045)	0.0132*** (0.0043)	0.0130*** (0.0043)	0.0020 (0.0017)	−0.0018 (0.0026)	−0.0020 (0.0025)	−0.0021 (0.0025)	0.1895 (0.1934)
Car manufacturer	−1.0512** (0.4600)	−1.0382** (0.4464)	−1.0240** (0.4433)	−0.4593** (0.2036)	−0.4634 (0.3028)	−0.4885* (0.2964)	−0.4877* (0.2934)	0.1895 (0.1934)
Chemical industry	−0.3888 (0.4320)	−0.4133 (0.4204)	−0.4108 (0.4174)	0.5312 (0.1901)	−0.1405 (0.2753)	−0.1370 (0.2738)	−0.1406 (0.2709)	0.5733*** (0.1710)
Europe	0.4007 (0.5290)	0.3972 (0.5103)	0.3707 (0.5051)	−2.3393*** (0.2109)	0.4141 (0.3043)	0.4212 (0.2971)	0.4189 (0.2932)	−0.2341 (0.1754)
US	0.1631 (0.4014)	0.2099 (0.3927)	0.2115 (0.3910)	−0.3354* (0.1929)	−0.1013 (0.2698)	−0.0269 (0.2671)	−0.0107 (0.2648)	0.5795*** (0.1658)
Constant	−4.2438*** (0.5790)	−3.5763*** (0.5775)	−3.8976*** (0.5816)	−0.3618 (0.6353)	−1.1890** (0.5252)	−2.3858*** (0.7008)	−3.730*** (0.6277)	−2.8050*** (0.4571)
Number of firms	85	85	85	85	85	85	85	
Number of firms-years	762	762	762	762	762	762	762	
Log-likelihood	−5254.7	−5127.1	−5119.9		−3174.4	−3055.3	−3033.6	−3022.5
Wald-Chi ²	2416.2	2668.2	2688.5	495.4	416.8	452.9	473.1	124.1
Change in Wald-Chi ²		252.0***	20.3			36.1**	20.2	

Notes: Standard error between brackets. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

^a Cognitive distance has been divided by 100 in the respective explanatory variables and interaction terms in order to have readable coefficients.

^b Year dummy variable-coefficients are included in the regressions but are not reported in the table.

shaped function of cognitive distance between alliance partners in the case of explorative patents: the linear term of cognitive distance has a (strongly significant) positive effect and the quadratic term a (strongly significant) negative effect. We did not find an inverted-U shaped function for exploitative patents. There is no relationship between cognitive distance and innovative performance in terms of “exploitative patents”—the coefficients of the linear and quadratic term are not significantly different from zero. While increasing cognitive distance has no effect on the innovative performance in the case of exploitative patents, it has an inverted-U shaped effect on explorative learning. The optimal cognitive distance at the mean value of technological capital is 38.4 on a scale between zero and hundred. Hence, *Hypothesis 2* is corroborated.

4.2. Technological capital

Model 2 tests Eq. (7) where absorptive capacity is not only a function of cognitive distance but also of the technological capital (TC) of an innovating firm, as has been specified in Eq. (6). Concerning the effects of technological capital in exploration and in exploitation (*Hypothesis 3*), we find a positive and significant effect for technological capital (prior cumulative patents) and a negative impact of technological capital in interaction with cognitive distance ($CD \times TC$). The introduction of these two variables also mitigates the (negative) effect of cognitive distance on innovativeness in the case of exploitative patents.

The results for technological capital are in line with the hypotheses, but we find a significant negative effect for the interaction term of TC and CD while we expect a positive effect. This is the case for both exploitative learning and explorative learning. In both cases, the existing technological capital of a firm becomes a burden at (high) cognitive distance levels: For both exploitation and exploration the effect of technological capital on innovative performance remains positive although the impact of technological capital is tempered with increasing cognitive distance. This is of course in contrast with what we expect from theory: firms with a large patent portfolio (technological capital) should have fewer problems in dealing with larger technological distances.

4.3. Control variables

Most control variables have the expected effect on the exploitative and explorative learning. Size has a positive and significant effect on the rate of innovation both for exploiting current technologies as well as for exploring

new technological areas. Since this explanatory variable is in the log form, its coefficient in the Poisson specification can be interpreted as elasticity between firm size and the dependent variables. The coefficients are in both tables smaller than one suggesting – all else equal – that the frequency of patenting increases with firm size but less than proportionately. As a result, small firms are more innovative than their larger counterparts. Interestingly, the coefficient for ‘firm size’ in the exploitation of current technologies is significantly larger than in the case of the exploration of new technologies. This difference shows that the small firms have a disproportionately large share of explorative patents. This is in line with the results of previous research that show that new and more radical inventions are likely to originate with entrants rather than incumbents (Cooper and Schendel, 1976; Foster, 1986).

As expected, R&D-intensity has a positive and significant effect on the innovation rate of the companies in the sample, and its impact is significantly larger in the case of exploitative learning compared to explorative learning. This strong impact on exploitative patents indicates that an increase in R&D-efforts will lead to more patents in the patent classes that the firm already masters. This is less the case for exploration of new technologies reflecting the high levels of uncertainty in explorative research.

As expected, age of the firm has a positive effect on exploitative patents and a negative, though not significant, effect on exploratory patents. This result suggests that established companies that had time to develop capabilities in particular technological fields have a clear competitive advantage over new entrants that still have to develop this technological expertise. By contrast, the negative coefficient for ‘explorative patents’ indicates that newly established firms might have a slight advantage in exploring new technological fields (although we have no conclusive evidence). This is in line with previous research that focused on the role of new firms in the creation of new technologies (Methe et al., 1997; Sorenson and Stuart, 2000).

To check for inter-industry differences in the propensity to patent (both explorative and exploitative) we introduced two dummy variables – car manufacturer and chemical industry, while the pharmaceutical industry is the default – to control for these differences. The coefficients for the car manufacturing industry are negative and statistically significant, suggesting that pharmaceutical companies (default) are more inclined to patent in both existing and new patent classes than firms in the two other industries. There are no significant differences between the chemical and the pharmaceutical

industry. The dummy variables indicating in which economic block the companies are based (America, Europe and the Middle-East, or the Far East) have no significant coefficients. Most year dummy variables have significant effects (not reported in the tables) indicating that the propensity to patent – both in existing and new patent classes – is variable over time.

5. Discussion

5.1. Anomaly

The erosion of the positive effect of technological capital on explorative learning and the possibility of a negative effect is at odds with [Hypothesis 3](#), where we argued that larger technological capital generally shows a better performance in dealing with cognitive distance. More specifically, we hypothesized that higher TC yields a general increase of absorptive capacity, in the form of an upward shift of the line that represents such capacity (see [Fig. 1](#)). However, according to the mathematical analysis that would yield a positive effect of the interaction term of TC and cognitive distance: see [Eq. \(7\)](#), but we observe a significant negative effect. Further inspection of the mathematics, elaborated in [Appendix B](#), indicates that there are three possible explanations for the observed negative effect of the interaction variable $TC \times CD$, as follows:

- (1) Technological capital not only increases the intercept of the line for absorptive capacity, but also its (negative) slope, in other words the marginal disutility of cognitive distance with regard to absorptive capacity. This would imply that a broader scope of technological capability becomes a liability for absorption at large cognitive distance. A wider field of competence makes it more difficult to understand something that is radically different. Along these lines, several authors ([Ahuja and Lampert, 2001](#); [Helfat, 1994](#); [Martin and Mitchell, 1998](#); [Nelson and Winter, 1982](#); [Stuart and Podolny, 1996](#)) have argued that these organizations have a propensity to adopt solutions in the neighborhood of existing solutions. Organizational routines and bureaucratic convenience favor projects that look for new solutions near old solutions over projects that rely on completely new principles. Resource allocation also follows established norms, controls, and procedures so that projects that build on antecedents are likely to be more easily selected and financed. This is rational when there is a complex system of many tightly coupled elements, because then the incorporation of

a novel element may upset the systemic integrity of the system as a whole ([Nooteboom, 2000](#)). Mathematical elaboration, given in [Appendix B](#), shows that this would yield a second interaction term, of technological capital and the square of cognitive distance ($TC \times CD^2$), with a negative effect.

- (2) Technological capital (also) yields an upward shift of the line representing the novelty value of cognitive distance. This reflects the idea that a broad technological basis yields better opportunities for Schumpeterian novel combinations: whatever a partner has to offer, has a larger chance of yielding a connection with something within the scope of technological capability. [Appendix B](#) shows that this yields no effect of the interaction term $TC \times CD^2$.
- (3) Technological capital (also) yields a smaller slope of the line of novelty value, in other words a smaller marginal utility of distance for novelty value. This implies a principle of decreasing returns to knowledge: with a broader technological basis one requires a larger distance to generate additional novelty value. In other words, this may be called the ‘boredom hypothesis’: the more one already knows, the further afield one has to go, in more exotic relations, to still find something new. [Appendix B](#) shows that this yields a positive effect of the interaction term $TC \times CD^2$.

In sum, addition of the interaction term of technological capital and quadratic cognitive distance allows for effects of technological capital on the marginal (dis)utilities of cognitive distance for absorptive capacity and novelty value.

5.2. Extended model

We can now discriminate between these three alternative explanations on the basis of the significance and the sign of an effect of the new interaction term $TC \times CD^2$. The results of the corresponding test are given in Model 3 of [Table 3](#) and are illustrated in [Figs. 2–4](#). The coefficient of the variable $TC \times CD^2$ is in both cases positive and significantly different from zero. These results support the third explanation: the idea that technological capital might reduce the steepness of the line of novelty value as a function of cognitive distance. Larger cognitive distance adds proportionately less value for the firm, the larger its existing technological competencies. In addition, the coefficient of the explorative patents is significantly larger than for the exploitative patents ([Table 3](#)). This indicates that the tempering effect of the existing technological competence in companies is

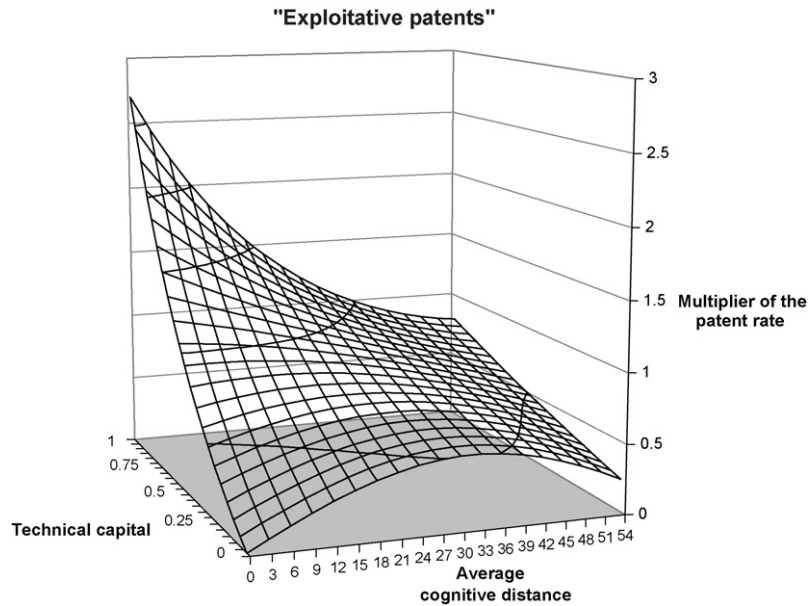


Fig. 2. Optimal cognitive distance for “exploitative patents” based on the results of the negative binomial regression in Table 3.

more drastic in the exploration of new fields than in the exploitation of technological fields where the innovating company has already build its strength.

However, as shown in Appendix B, if higher TC only reduces the slope of the line for novelty value, there would be no direct effect of TC on innovative performance, but we do observe that effect. To yield that empirical result, TC must also either increase the inter-

cept of the line for absorptive capacity, as originally hypothesized, or also increase the intercept of the line for novelty value, or both.

Note, finally, that with the extended model, including the added interaction term $TC \times CD^2$, allowing for effects of technological capital on marginal (dis)utilities of cognitive distance, all hypotheses concerning the effects of cognitive distance, linear, quadratic and in

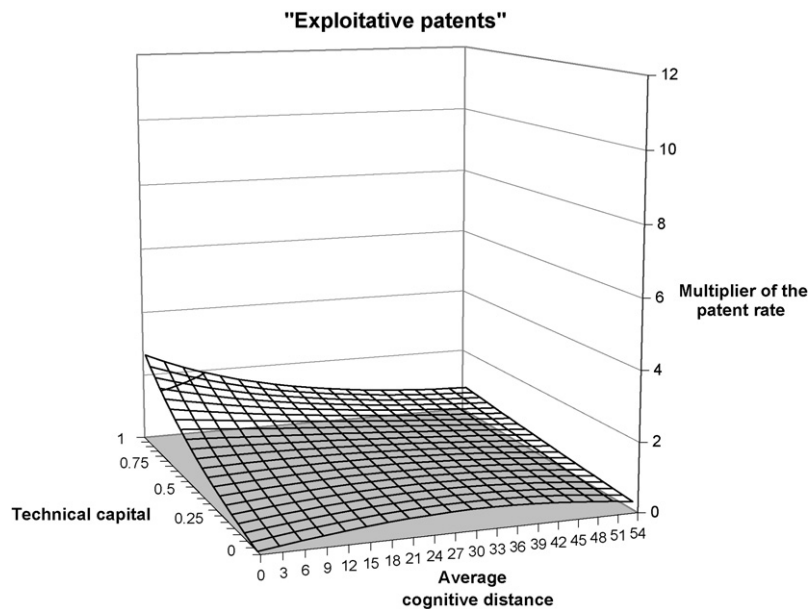


Fig. 3. Optimal cognitive distance for “exploitative patents” based on the results of the negative binomial regression in Table 3 but the axes are resized in the same way as in Fig. 4.

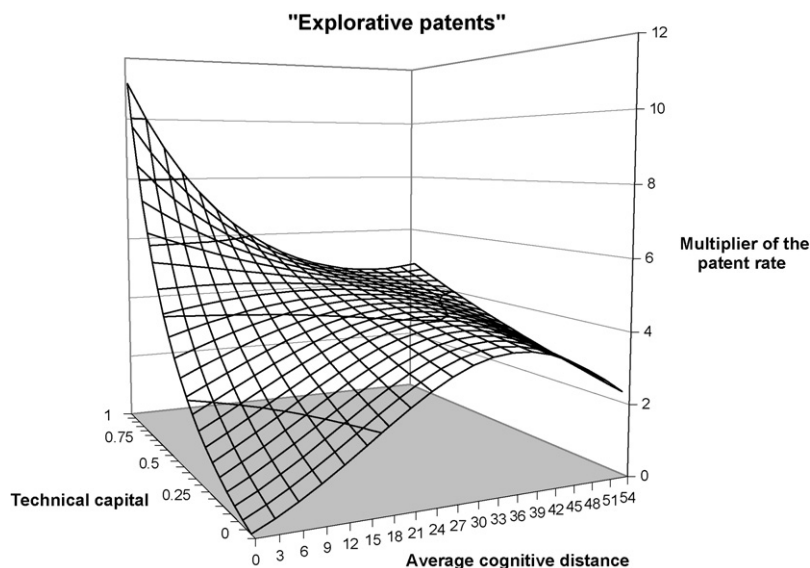


Fig. 4. Optimal cognitive distance for “explorative patents” based on the results of the negative binomial regression in Table 3.

interaction with cumulative patents, are now confirmed. They are not only confirmed for explorative performance but also for exploitative performance, though for the latter the effects are still much smaller than for the former, which is in accordance with our hypotheses (see Table 3, Model 3).

In order to show the robustness of our findings we also present the results of a random effect negative binomial regression. The results for model 3 are completely in line with those of the Poisson regression. The negative binomial results show a somewhat smaller difference between the impact of CD and TC on exploitation on the one hand and exploration on the other hand. However, the impact of these variables is (independent of the regression method) many times stronger for exploration than for exploitation as can be seen through the comparison of Figs. 2 and 3 with Fig. 4.

6. Conclusions

6.1. Key findings and implications

In this paper we have considered the relation between cognitive distance and innovation performance of firms cooperating in technology-based alliances. We have interpreted cognitive distance in terms of differences in technological knowledge between firms. The key finding is that the hypothesis of an inverted U-shaped effect of cognitive distance on innovation performance of firms is confirmed. Moreover, as expected, we found that the positive effect for firms is much higher when engag-

ing in more radical, exploratory alliances than in more exploitative alliances.

The effect of cumulative R&D, expressed in the scope of patents, turns out to be mixed. It may increase absorptive capacity, as originally hypothesized, and it may also increase the level of novelty value (intercept of the line), but there is clear evidence that it also reduces the effect of cognitive distance on novelty value (slope of the line of novelty value).

The implications are important, for theory and practice. Rather than assuming that differences in cognition (in terms of technological knowledge) only complicate collaboration, one should also recognize the positive potential of such differences.

A consequence for firms is that they need to be aware that in cooperating with others in alliances there is a trade-off to be made between the opportunity of novelty value and the risk of misunderstanding. This is an important insight that complements findings reported in some recent studies. In this recent literature, the benefits of searching for and accessing heterogeneous resources have been stressed considerably (Ahuja and Katila, 2004; Rosenkopf and Nerkar, 2001; Rosenkopf and Almeida, 2003), apparently driven by the (implicit) goal to benefit from a rise in novelty value when cognitive distance increases. However, this clearly ignores the notion that employing such strategies comes at a risk of decreasing understanding with a negative effect on innovation performance. Whereas in the literature on international business, indicated earlier, the positive effect of distance is neglected, this literature runs the risk of neglecting

its problems. To deal with this, firms should seek partners that are at an optimal cognitive distance. They may assess this in ways similar to those employed here: by inspecting their patent portfolio.

Such optimal distance is not fixed, but depends on one's past investment in building technological knowledge as a basis of absorptive capacity. The mixed effect of cumulative R&D suggests that while there may be increasing returns in absorptive capacity, improving the general ability to understand and appreciate novelty value in collaboration, there are decreasing returns to knowledge in finding further novelty: the more one knows the further away one has to look for novelty.

Additional findings are that, as expected from earlier studies of innovation and firm size, large firm advantage is less in more radical, exploratory innovation than in more exploitative innovation. Also, as expected, older firms perform better in more exploitative alliances but not better, and perhaps worse, in exploratory alliances. Finally, in the absence of any remaining effects from different regions (US, EU), it appears that any regional differences are captured in the explanatory variables.

6.2. Limitations and future directions

One limitation of the present study is that by taking firms as the unit of analysis, we had to average cognitive distance across the various alliance partners of the firm. An alternative is to take alliances as the unit of analysis, with their individual results in terms of innovation. For this one can no longer use patents as a measure of innovative performance (since they are given on the firm level and cannot without additional data be allocated to individual alliances)

The present study does not include a test of the derived hypothesis that ongoing duration of an alliance reduces cognitive distance, yielding an inverted-U shaped effect of duration on innovative performance, which was confirmed by Wuyts et al. (2005). With the data used here, no reliable measure of alliance duration could be constructed.

Apart from the limitations of using patents, widely discussed in the literature, as a proxy for technological capital, we face another limitation in the dichotomization of explorative and exploitative patents. This dichotomization in two categories could be removed by introducing an interval variable representing the degree of exploration in the new patents of a company in a particular year. The information entailed in this type of variable is richer than the two categories we used now.

Most authors have used *patent citations* as the basis for the calculation exploration/exploitation variables:

Katila and Ahuja (2002), Nerkar (2003) and Rosenkopf and Nerkar (2001). We could trace one (unpublished) paper by Riitta Katila that takes the same approach as ours – *patent classes* – to distinguish between exploitation and exploration. Future research on the issue of exploration versus exploitation should aim to combine the two measures to get a more detailed measurement of exploration: is exploration (no citations to own patents) in a firm's existing technology classes different from that in new (explorative) classes?

It would be interesting to further test the indication found in this study that experience in R&D reduces the marginal novelty from cognitive distance, and to try and answer the question, left undecided by our empirical results, whether in addition such experience increases both the level of absorptive capacity and the novelty value, or only one of them.

Next, in this paper we only elaborated on the cognitive distance between the focal firm and its alliance partners, not on the cognitive distance among the alliance partners themselves. An interesting direction for future research is to analyze how the average cognitive distance between the alliance partners might affect the impact of alliance portfolio's on the innovativeness of companies, both in terms of exploitation and exploration.

Another limitation is related to the concept of absorptive capacity. Cohen and Levinthal (1990) underlined that this notion not only refers to the acquisition or assimilation of information by an organization, but also on the organization's ability to exploit it⁴; An organization's absorptive capacity not only depends on the organization's direct interface with the external environment, but also on transfers of knowledge within and across subunits in the organization. As a result, to understand a firm's absorptive capacity, one should focus on the structure of communication between the external environment and the organization, as well as among the internal innovation strategy and organization of the R&D that determine the character and distribution of technological knowledge within the organization (Levinthal and March, 1993; Argyres and Silverman, 2004). This dimension of absorptive capacity is not taken into account as the current dataset did not provide information about the focal firms' innovation strategy and organization. This is an avenue for future research which some scholars recently started to explore (Bamford et al., 2003; Foss et al., 2005; Zhang et al., 2006).

A more fundamental challenge is the following. In the underlying theory of knowledge, cognition is assumed

⁴ We thank an anonymous referee for mentioning this.

to be a broad concept, including both cognition in the narrow sense of intellect, in perception and inference, and cognition in the wider sense of reflexive, normative and emotional behavior, in perception, attribution, and value judgments. In the present study we considered only cognitive distance in a limited sense of differences in technological knowledge, inferred from patent data. In future research it would be worthwhile to separate out differences in different dimensions of cognition. The hypothesis for such a study would be that for optimal collaboration distance should be relatively small in cognition concerning governance, i.e. in ways and styles of dealing with relational risk, in more or less inclination to trust, for example, and should be relatively large in substantive aspects of intellectual knowledge. Wuyts et al. (2005) give an example of a study where a distinction was made between difference in technological competence and distance in organizational variables.

An obvious extension of this study is to focus on the role of an alliance network in terms of the technological distance between partners, a firm's network position (centrality) and total network density. A subsequent study could show how these three elements of an alliance network, apart and in combination, affect the 'twin tasks' in exploration, namely novelty creation on the one hand and its efficient absorption on the other hand.

Appendix A. Effect of the slope of the line for novelty value (b_2)

For the effect of b_2 on optimal cognitive distance (CD^*) and corresponding innovative performance (L^*) we consider their derivatives with respect to b_2 :

$$\frac{dCD^*}{db_2} = \frac{1}{2} \frac{b_1}{b_2^2} > 0 \quad (A1)$$

This proves that an increase of b_2 yields an increase of optimal cognitive distance.

$$\frac{dL^*}{db_2} = \frac{(a_1b_2 - b_1a_2)(a_1b_2 + b_1a_2)}{4a_2b_2^2} > 0 \quad (A2)$$

if $a_1 > b_1a_2/b_2$, and this was assumed earlier.

This proves that an increase of b_2 yields an increase of optimal innovative performance.

Appendix B. Negative effect of interaction $TC \times CD$

B.1. Alternative 1

Higher TC increases the negative slope of the line for absorptive capacity (a_2)

$$\text{Let } a_2 = d_1 + d_2TC$$

Substitution into the formula (3) for innovative performance L then yields:

$$L = a_1b_1 + (a_1b_2 - b_1d_1)CD - b_1d_2TC \times CD - b_2d_1CD^2 - b_2d_2TC \times CD^2 \quad (B1)$$

Here there is a negative effect of the interaction variable $TC \times CD$ and a *negative* effect of the interaction variable $TC \times CD^2$. Note that there is no effect of TC separately.

B.2. Alternative 2

Higher TC increases the intercept of the line for novelty value (b_1)

$$\text{Let } b_1 = e_1 + e_2TC$$

Substitution into (3) yields:

$$L = a_1e_1 + a_1e_2TC + (a_1b_2 - a_2e_1)CD - a_2e_2TC \times CD - a_2b_2CD^2 \quad (B2)$$

Here there is also a negative effect of interaction $TC \times CD$, *no* effect of interaction $TC \times CD^2$, and there is a positive effect of TC directly.

B.3. Alternative 3

Higher TC decreases the positive slope of the line for novelty value (b_2)

$$\text{Let } b_2 = f_1 - f_2TC$$

Substitution into (3) yields:

$$L = a_1b_1 + (a_1f_1 - b_1a_2)CD - a_1f_2TC \times CD - a_2f_1CD^2 + a_2f_2TC \times CD^2 \quad (B3)$$

Here there is also a negative effect of the interaction $TC \times CD$, a *positive* effect of interaction $TC \times CD^2$, and no direct effect of TC.

For a positive effect of TC on the intercept of the line for absorptive capacity (a_1), assuming $a_1 = c_1 + c_2TC$, we found, in (7),

$$L = c_1b_1 + c_2b_1TC + (c_1b_2 - b_1a_2)CD + c_2b_2TC \times CD - a_2b_2CD^2 \quad (B4)$$

To reproduce the empirical results of a positive direct effect of TC, a positive effect of CD, a negative effect of CD^2 , a negative effect of interaction $TC \times CD$, and a positive effect of interaction $TC \times CD^2$, we need a combination of (B3) and (B2) and/or (B4). In other words, TC has a negative effect on the positive slope of the line for novelty value, and a positive effect on either the intercept of the line for absorptive capacity or the intercept of the line for novelty value, or both.

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